

Extended Materials

for

SPSID: A Single-Parameter Shrinkage Inverse-Diffusion for Denoising Gene-Regulatory Networks

This document provides additional implementation details, extended experiments, and extra figures/tables accompanying the manuscript *SPSID: A Single-Parameter Shrinkage Inverse-Diffusion for Denoising Gene-Regulatory Networks*. These materials are provided in the project repository.

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1 Additional Methodological Details

Algorithm A1 SPSID workflow

Require: Raw TF–target matrix $W_{\text{obs}} \in \mathbb{R}^{m \times n}$.

Ensure: Denoised matrix $W_{\text{dir}} \in \mathbb{R}^{m \times n}$.

Step 1: Pre-processing

- 1: Padding to square $n \times n$

$$W_{\text{pad}} \leftarrow \begin{bmatrix} W_{\text{obs}} & \\ \mathbf{0}_{(n-m) \times m} & I_{n-m} \end{bmatrix} \in \mathbb{R}^{n \times n}$$

▷ add self-loops for pure targets.

- 2: ε -perturbation

$$\widetilde{W} \leftarrow W_{\text{pad}} + \varepsilon_1 \mathbf{J} + \varepsilon_2 \mathbf{I}, \quad \varepsilon_1 = \varepsilon_2 = 10^{-6}$$

▷ ensure irreducibility and aperiodicity.

- 3: Row-stochastic normalisation

$$\mathbf{D} \leftarrow \text{diag}(\widetilde{W}\mathbf{1}), \quad \mathbf{P}_{\text{obs}} \leftarrow \mathbf{D}^{-1}\widetilde{W}$$

▷ create transition matrix.

Compute stationary distribution π of $\mathbf{P}_{\text{obs}}$.

Step 2: Denoising

- 4: Shrinkage inverse diffusion

$$\mathbf{P}_{\text{dir}} \leftarrow \mathbf{P}_{\text{obs}}(I + \mathbf{P}_{\text{obs}} + \lambda I)^{-1}, \quad \lambda \leftarrow 10^3$$

▷ remove indirect paths while stabilising the inverse.

Step 3: Re-weighting

- 5: Node-specific scaling

$$W_{\text{dir}} \leftarrow \text{diag}(\pi) \mathbf{P}_{\text{dir}} \in \mathbb{R}^{n \times n}$$

▷ restore biological context.

- 6: **return** $W_{\text{dir}}^{\text{TF}} \leftarrow W_{\text{dir}}[1:m, 1:n]$

▷ keep the first m TF rows (shape $m \times n$).

2 Additional Simulation Results

Additional Table A1: Mean AUROC and AUPR under varying sparsity (ρ), noise (σ), and transitivity (β).

(a) Varying network density ($\sigma = 0.5, \beta = 0.5$)										
Method	$\rho = 0.01$		$\rho = 0.03$		$\rho = 0.05$		$\rho = 0.07$		$\rho = 0.09$	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
SPSID	0.909	0.177	0.896	0.277	0.878	0.305	0.856	0.313	0.832	0.317
RENDOR	0.865	0.096	0.842	0.170	0.816	0.203	0.789	0.222	0.764	0.238
ND	0.857	0.105	0.835	0.164	0.807	0.187	0.778	0.201	0.751	0.214
NE	0.830	0.089	0.810	0.168	0.777	0.199	0.744	0.216	0.717	0.230
Silencer	0.526	0.012	0.521	0.033	0.517	0.054	0.514	0.074	0.511	0.094
ICM	0.453	0.009	0.460	0.027	0.467	0.046	0.472	0.064	0.476	0.084

(b) Varying noise level ($\rho = 0.05, \beta = 0.5$)										
Method	$\sigma = 0.1$		$\sigma = 0.3$		$\sigma = 0.5$		$\sigma = 0.7$		$\sigma = 0.9$	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
SPSID	0.957	0.528	0.940	0.454	0.878	0.303	0.811	0.208	0.758	0.156
RENDOR	0.939	0.379	0.894	0.296	0.816	0.202	0.750	0.146	0.703	0.116
ND	0.925	0.332	0.887	0.270	0.807	0.187	0.740	0.135	0.693	0.108
NE	0.884	0.311	0.853	0.278	0.777	0.199	0.715	0.144	0.671	0.114
Silencer	0.537	0.059	0.523	0.055	0.516	0.054	0.514	0.053	0.511	0.052
ICM	0.439	0.042	0.455	0.044	0.466	0.045	0.473	0.046	0.478	0.047

(c) Varying transitive strength ($\rho = 0.05, \sigma = 0.5$)										
Method	$\beta = 0.1$		$\beta = 0.3$		$\beta = 0.5$		$\beta = 0.7$		$\beta = 0.9$	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
SPSID	0.909	0.446	0.897	0.381	0.879	0.304	0.858	0.237	0.837	0.188
RENDOR	0.855	0.269	0.839	0.240	0.816	0.201	0.792	0.166	0.765	0.136
ND	0.847	0.251	0.831	0.226	0.807	0.187	0.780	0.150	0.752	0.123
NE	0.804	0.245	0.796	0.227	0.777	0.199	0.754	0.168	0.725	0.141
Silencer	0.525	0.054	0.522	0.054	0.518	0.054	0.513	0.053	0.510	0.053
ICM	0.460	0.045	0.460	0.045	0.466	0.045	0.471	0.046	0.478	0.047

Note. Bold values indicate the best results. Methods shown are SP SID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).

Additional Table A2: Mean AUROC and AUPR on the G2 simulation model at the boundary condition ($\beta_1 = 1.0$).

Boundary condition simulation ($\rho = 0.05, \sigma = 0.5, \beta_1 = 1.0$)		
Method	AUROC	AUPR
SPSID	0.828	0.172
RENDOR	0.754	0.125
ND	0.740	0.113
NE	0.714	0.130
Silencer	0.508	0.052
ICM	0.481	0.047

Note. Bold values indicate the best results. Methods shown are SP SID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).

Additional Table A3: Mean AUROC and AUPR on the higher-order simulation model.

Varying cubic (G3) transitive strength (β_2) ($\rho = 0.05$, $\sigma = 0.5$, $\beta_1 = 0.5$)										
Method	$\beta_2 = 0.1$		$\beta_2 = 0.3$		$\beta_2 = 0.5$		$\beta_2 = 0.7$		$\beta_2 = 0.9$	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
SPSID	0.869	0.271	0.829	0.202	0.788	0.157	0.757	0.131	0.733	0.115
RENDOR	0.814	0.201	0.785	0.178	0.750	0.147	0.719	0.123	0.694	0.108
ND	0.805	0.184	0.775	0.152	0.736	0.123	0.705	0.105	0.679	0.094
NE	0.764	0.189	0.727	0.158	0.693	0.133	0.667	0.115	0.649	0.103
Silencer	0.517	0.054	0.512	0.054	0.508	0.053	0.506	0.053	0.505	0.052
ICM	0.467	0.045	0.475	0.046	0.480	0.047	0.485	0.047	0.487	0.048

Note. Bold values indicate the best results. Methods shown are SP SID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).

Additional Table A4: SP SID sensitivity to shrinkage parameter λ

λ	AUROC			AUPR		
	Mean	Lower CI	Upper CI	Mean	Lower CI	Upper CI
100	0.878	0.878	0.879	0.304	0.303	0.305
200	0.879	0.878	0.879	0.304	0.303	0.305
300	0.878	0.878	0.879	0.304	0.303	0.306
400	0.879	0.878	0.879	0.305	0.303	0.306
500	0.879	0.878	0.879	0.305	0.304	0.306
600	0.878	0.878	0.879	0.304	0.302	0.305
700	0.878	0.878	0.879	0.304	0.303	0.305
800	0.878	0.878	0.879	0.304	0.303	0.305
900	0.878	0.878	0.879	0.304	0.303	0.305
1000	0.879	0.878	0.879	0.304	0.303	0.306
5000	0.878	0.878	0.879	0.304	0.303	0.305
10000	0.878	0.878	0.879	0.304	0.303	0.306

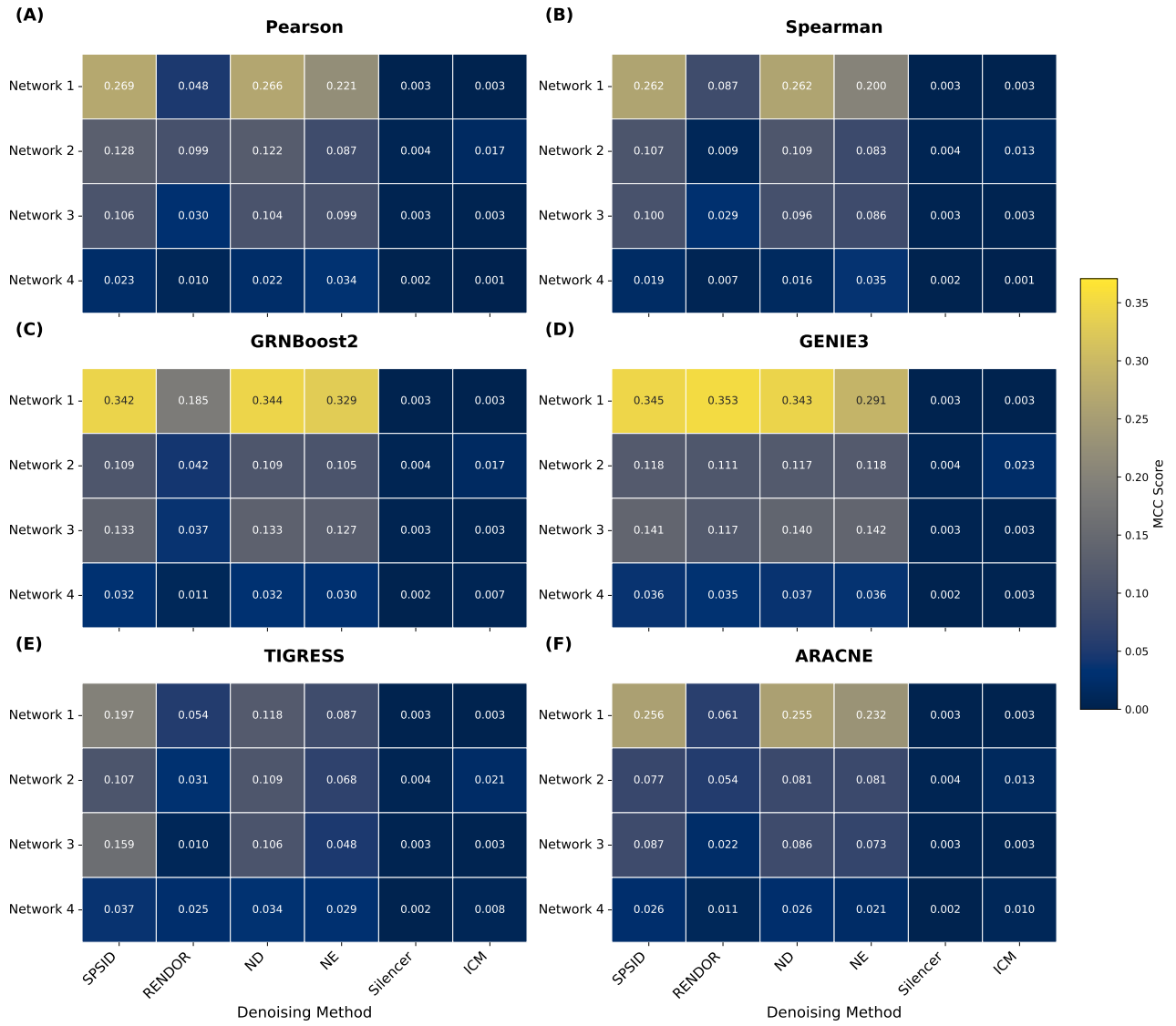
Note. Confidence intervals are ± 1.96 SE over 1000 replicates.

Additional Table A5: Runtime comparison on synthetic networks of increasing size n .

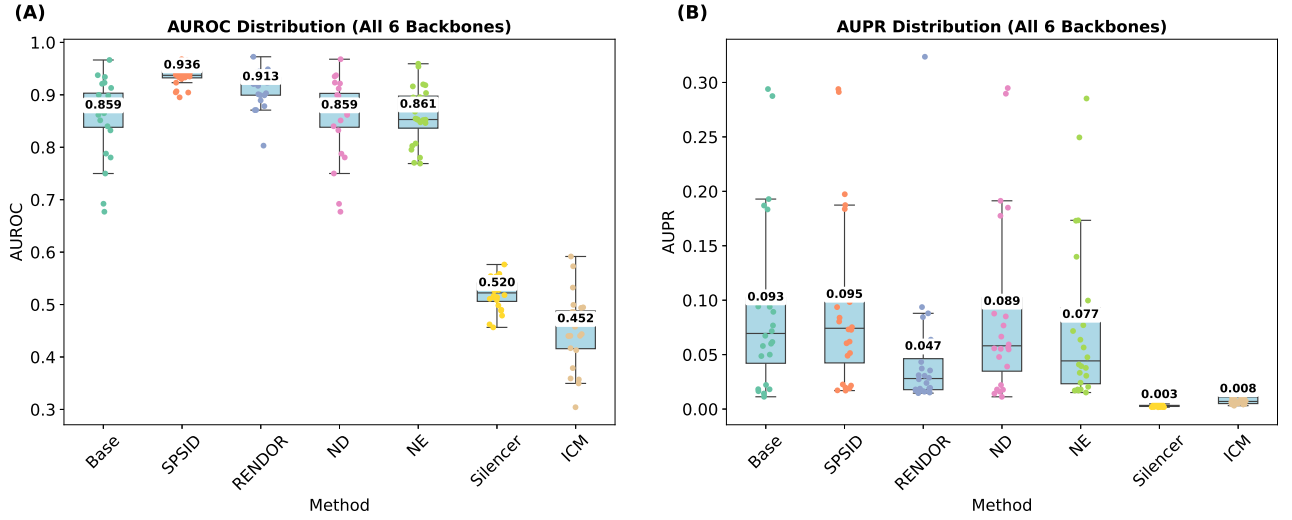
Method \ n	500	1000	2000	3000	4000	5000	6000
SPSID	0.24 $\pm$ 0.02	1.01 $\pm$ 0.02	5.33 $\pm$ 0.01	17.49 $\pm$ 0.17	48.47 $\pm$ 0.06	89.67 $\pm$ 0.14	148.72 $\pm$ 0.25
RENDOR	0.14 $\pm$ 0.03	0.53 $\pm$ 0.01	5.04 $\pm$ 0.16	18.48 $\pm$ 0.55	53.94 $\pm$ 0.21	102.03 $\pm$ 0.41	174.87 $\pm$ 1.24
ND	0.42 $\pm$ 0.00	3.38 $\pm$ 0.04	53.70 $\pm$ 0.32	246.55 $\pm$ 0.73	651.77 $\pm$ 1.24	1298.00 $\pm$ 2.70	2373.05 $\pm$ 1.05
NE	0.11 $\pm$ 0.00	0.48 $\pm$ 0.03	2.53 $\pm$ 0.02	10.38 $\pm$ 0.14	23.06 $\pm$ 0.62	42.23 $\pm$ 0.29	71.24 $\pm$ 0.16
Silencer	0.03 $\pm$ 0.01	0.17 $\pm$ 0.03	0.82 $\pm$ 0.02	3.07 $\pm$ 0.12	6.09 $\pm$ 0.14	10.59 $\pm$ 0.13	16.92 $\pm$ 0.23
ICM	0.03 $\pm$ 0.00	0.17 $\pm$ 0.07	0.58 $\pm$ 0.03	2.09 $\pm$ 0.17	3.86 $\pm$ 0.17	6.26 $\pm$ 0.06	10.16 $\pm$ 0.06

Notes: Entries report mean $\pm$ standard deviation of wall-clock time (in seconds) across repeated runs on the same hardware.

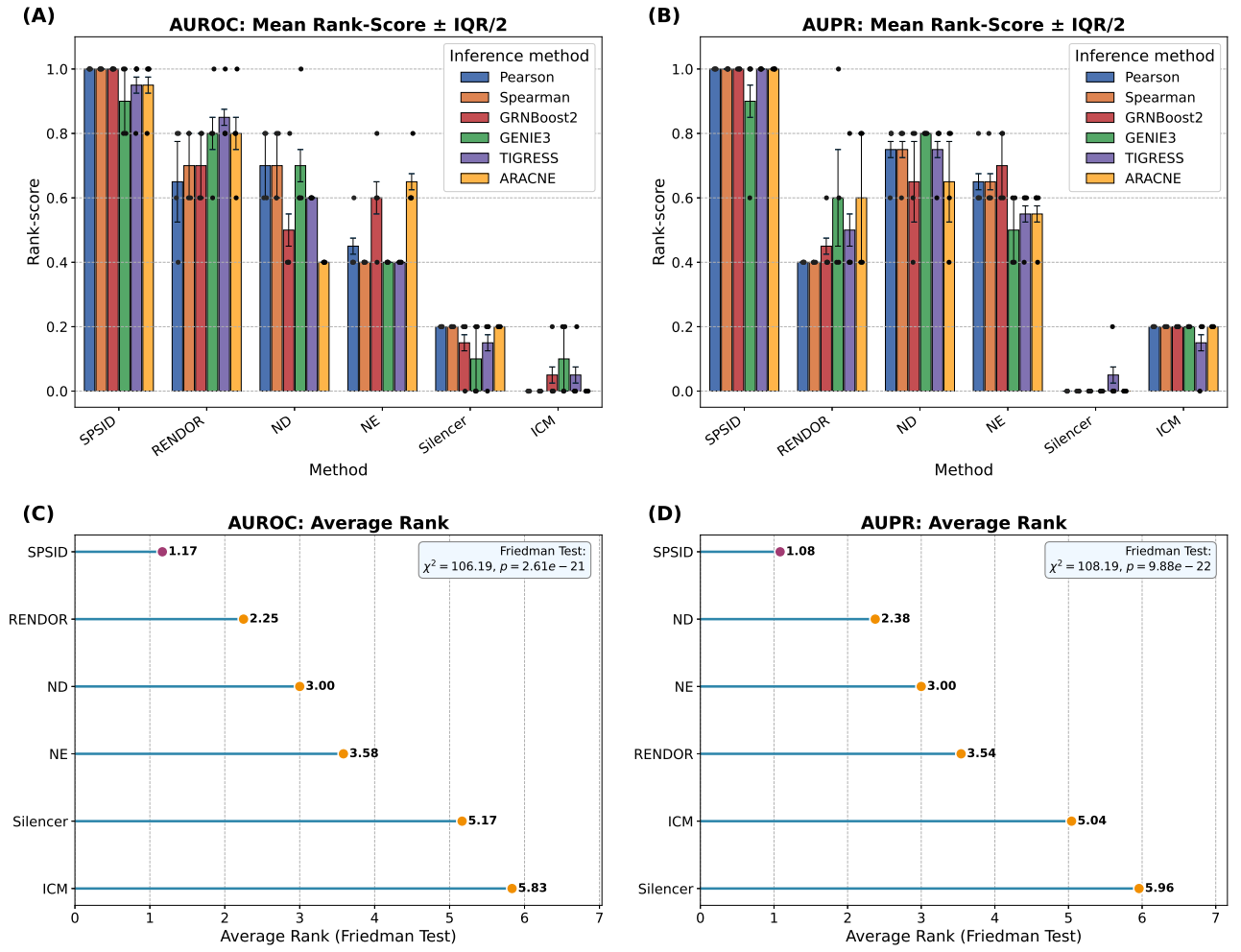
3 Additional DREAM5 Benchmark Results



Additional Figure A1: DREAM5 MCC benchmark performance. Facet heatmap displaying MCC scores. Each panel (A–F) represents one of the 6 upstream inference methods. Columns compare SPSID against 5 other denoising baselines across the 4 DREAM5 networks.



Additional Figure A2: Box plots detailing the performance distribution over the 24 DREAM5 combinations (4 networks $\times$ 6 upstream inference methods) for the base network and each denoising method. (A) AUROC distribution. (B) AUPR distribution. For each method, the light blue box indicates the interquartile range (IQR) from the 25th to the 75th percentile, with the median marked by a horizontal line inside. The whiskers extend to show the data range. Coloured dots represent the individual scores for each task, and the arithmetic mean is printed as a numeric value with a light background. Methods shown are SPSID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).



Additional Figure A3: A comprehensive analysis of method ranks over the 24 DREAM5 combinations. (A) and (B) Grouped bar charts display the mean rank-score for AUROC and AUPR, respectively. Error bars indicate half the inter-quartile range (IQR/2), providing a measure of variance across the six inference engines. (C) and (D) Lollipop plots show the average ordinal rank determined by the Friedman test for AUROC and AUPR. In all panels, superior performance is indicated by a higher rank-score (for the bar charts) or a lower average rank (for the lollipop plots). Methods shown are SPSID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).

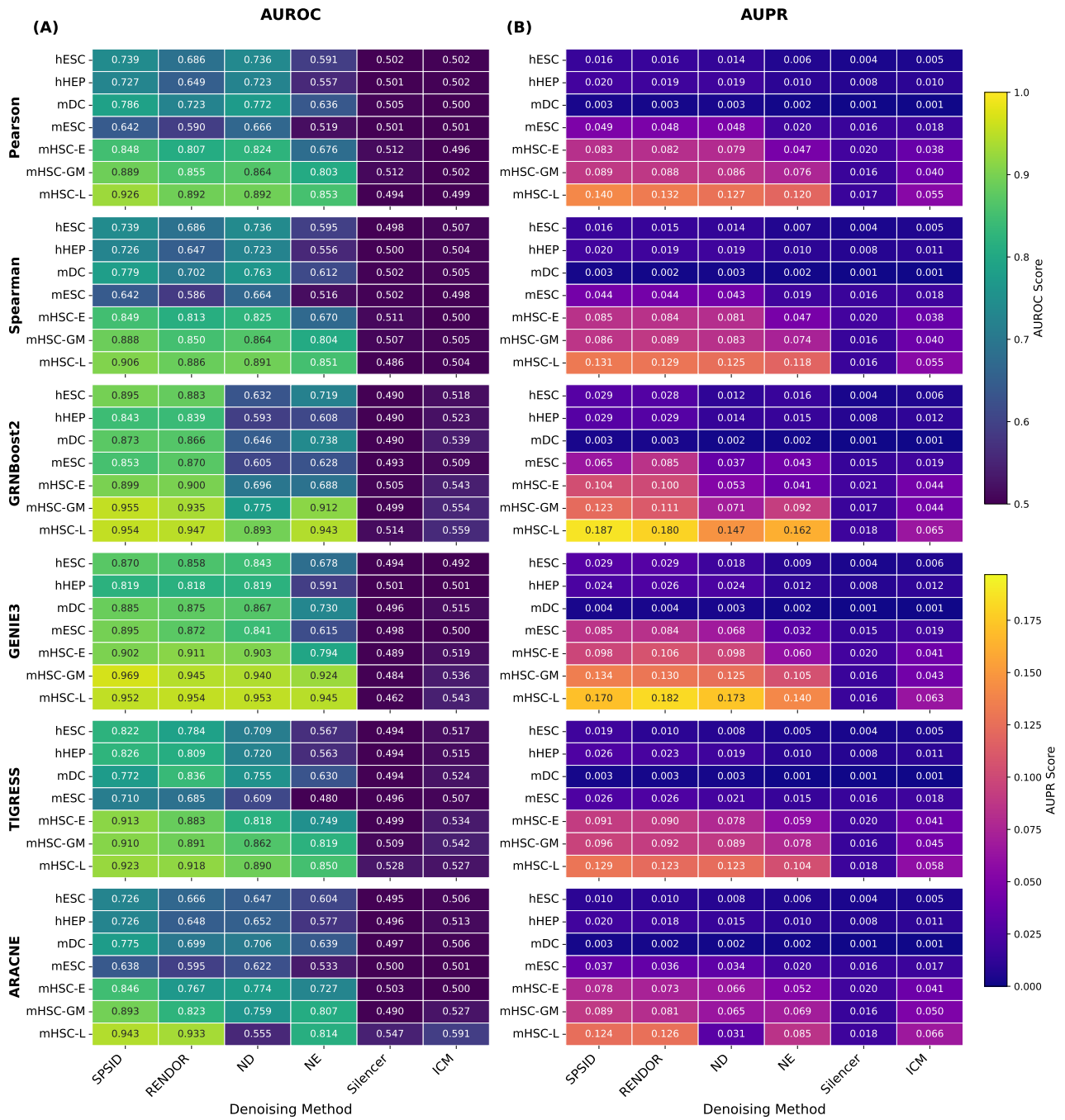
Additional Table A6: Percentage improvement for each DREAM5 network, reported as median and Interquartile Range (IQR).

Method	Network 1 (<i>in silico</i>)				Network 2 (<i>S. aureus</i>)			
	AUROC _{med}	AUROC _{IQR}	AUPR _{med}	AUPR _{IQR}	AUROC _{med}	AUROC _{IQR}	AUPR _{med}	AUPR _{IQR}
ICM	-61.57	4.72	-97.65	0.58	-44.84	3.91	-82.39	5.45
ND	0.04	0.05	-0.93	2.68	0.01	0.05	-1.87	6.26
NE	-2.06	9.06	-12.71	15.98	-8.10	4.46	-20.80	10.97
RENDOR	0.90	7.30	-73.20	12.68	7.32	4.41	-42.89	37.61
SPSID	2.16	5.35	0.21	0.96	9.53	3.69	4.73	5.46
Silencer	-40.12	5.09	-98.51	0.55	-36.33	2.39	-94.13	1.66

Method	Network 3 (<i>E. coli</i>)				Network 4 (<i>S. cerevisiae</i>)			
	AUROC _{med}	AUROC _{IQR}	AUPR _{med}	AUPR _{IQR}	AUROC _{med}	AUROC _{IQR}	AUPR _{med}	AUPR _{IQR}
ICM	-48.46	5.73	-90.15	2.28	-43.81	8.33	-50.63	7.40
ND	0.01	0.04	-4.43	8.27	-0.01	0.01	-1.15	1.24
NE	-2.61	9.59	-32.60	34.66	-2.17	4.98	1.36	10.67
RENDOR	0.78	12.88	-67.95	5.24	5.24	7.91	-3.77	21.25
SPSID	2.60	8.70	0.58	4.44	5.78	6.07	6.73	12.00
Silencer	-39.48	5.41	-95.01	1.91	-42.29	4.56	-89.64	2.59

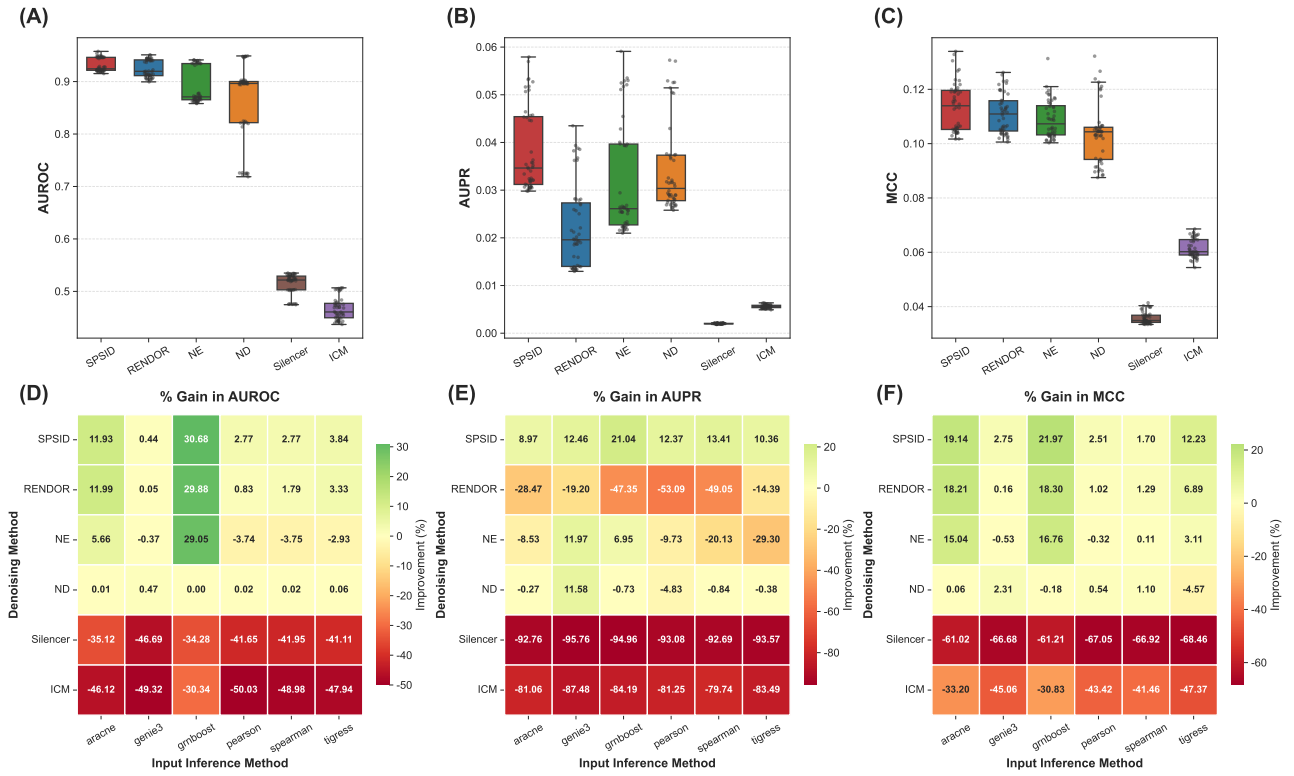
Note. Values are *median* and *Interquartile Range (IQR)*; IQR is the difference between the 75th and 25th percentiles (Q3 - Q1) over the six upstream inference methods. Methods shown are SPSID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).

4 Additional scRNA-seq Benchmark Results

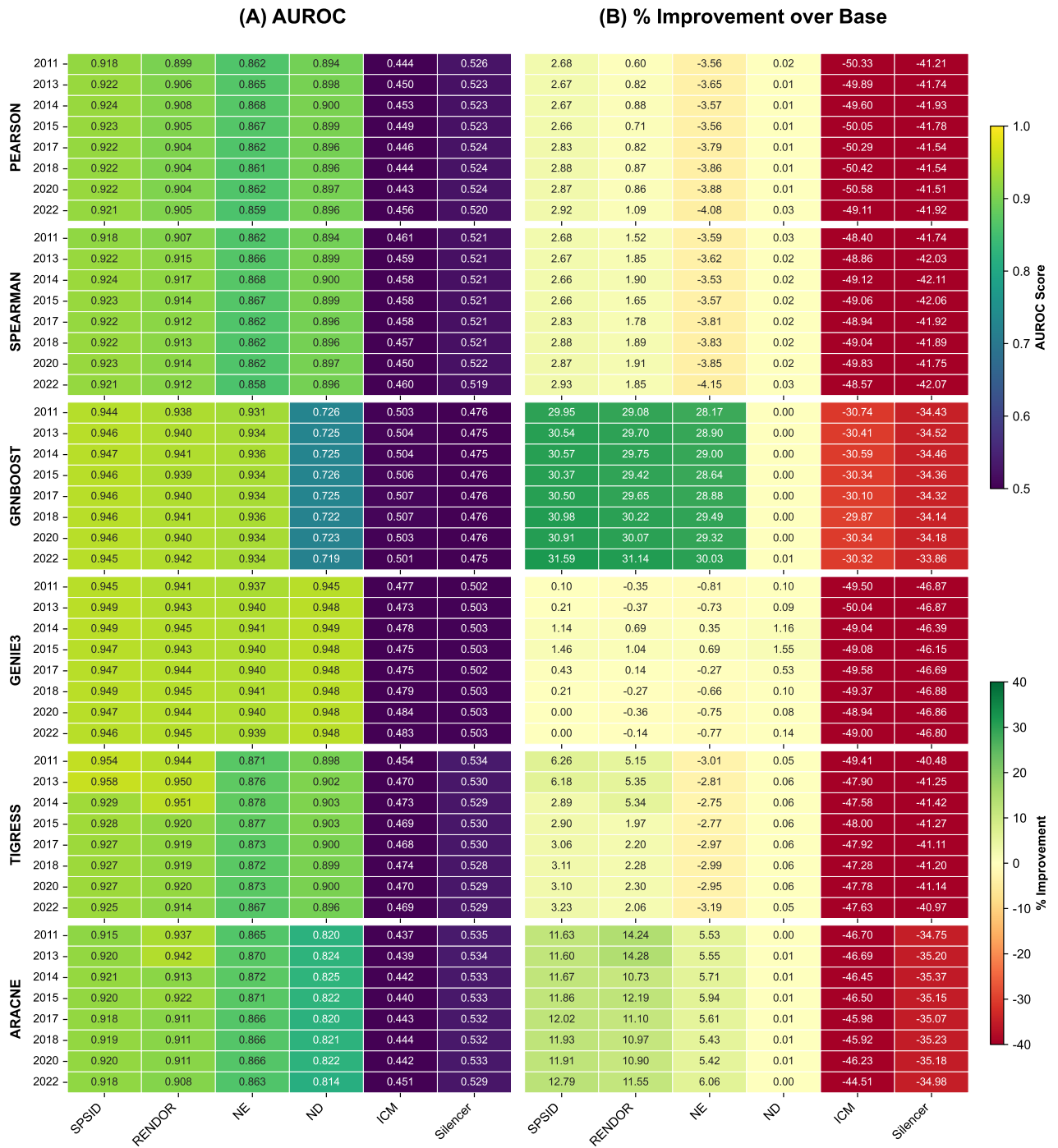


Additional Figure A4: scRNA-seq benchmark performance. Facet heatmap displaying AUROC (A) and AUPR (B) scores. Each row-group represents one of the 6 upstream inference methods, and each row within a group represents one of the 7 scRNA-seq datasets. Columns show the SPSID and 5 other denoising methods.

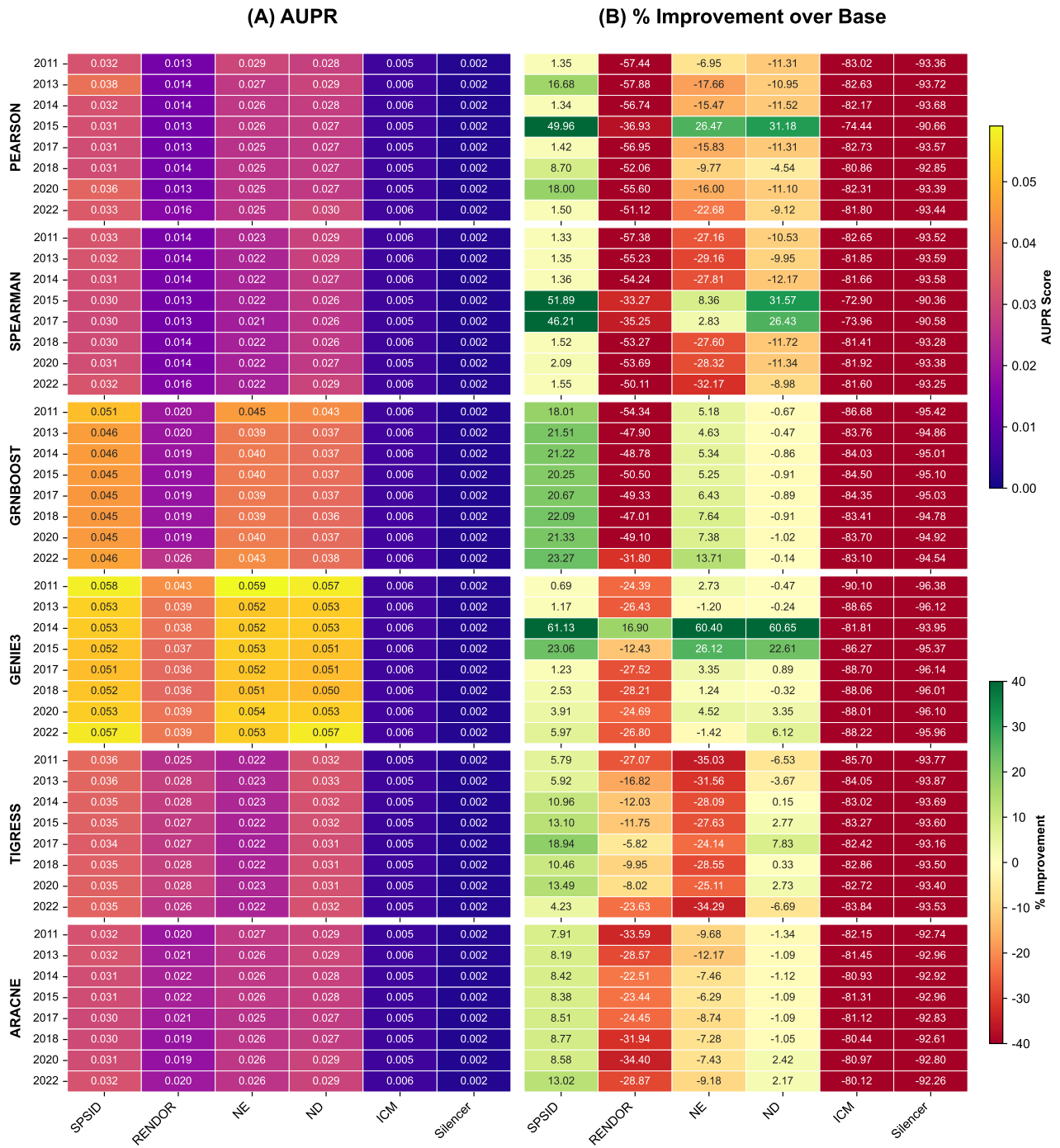
5 Additional Abasy Robustness Analysis



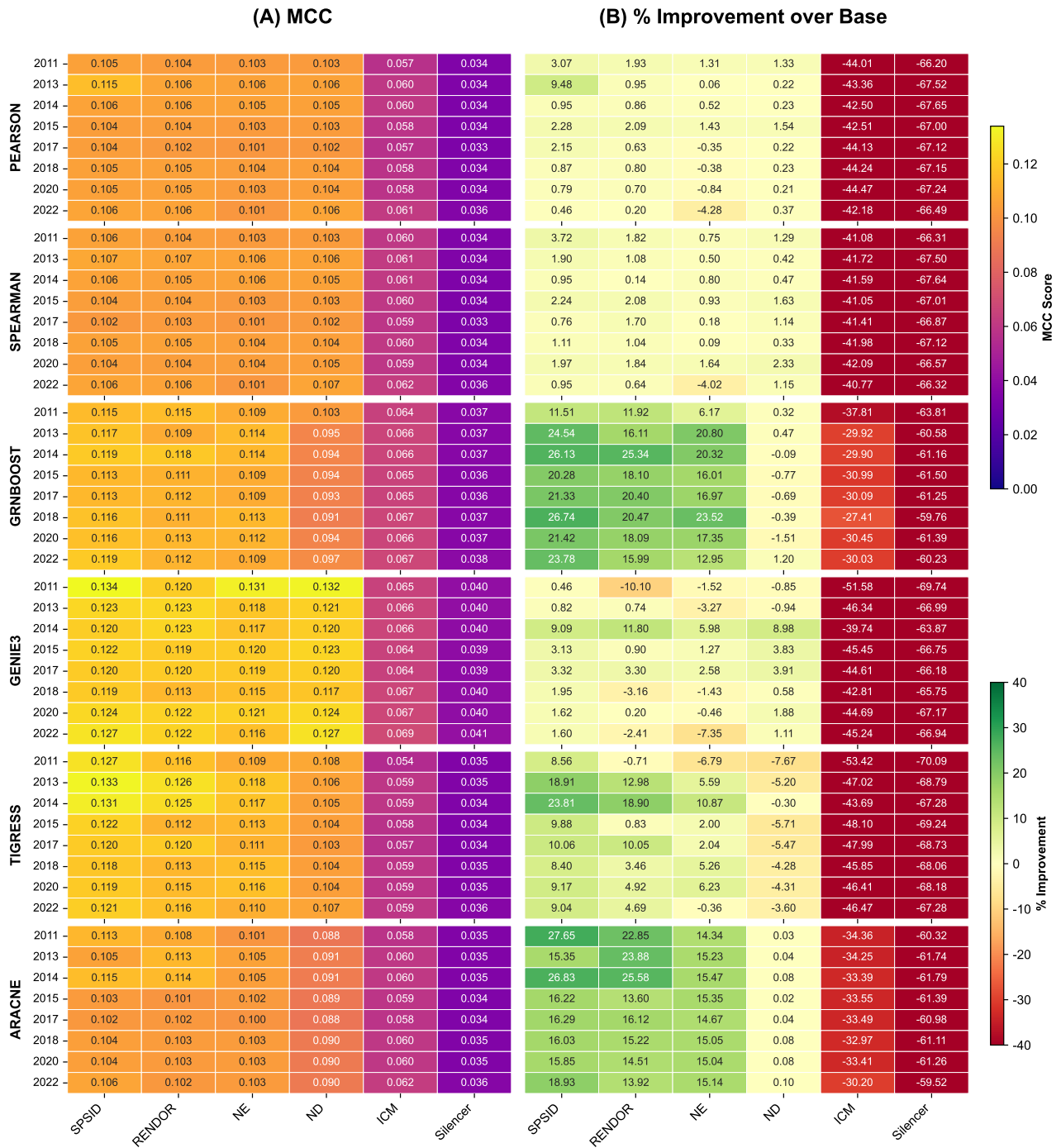
Additional Figure A5: Summary of Abasy (v2.9, strong) snapshot-based evaluations. (A–C) Box plots of AUROC, AUPR, and MCC across denoising methods. (D–F) Average percentage gain over the raw (Base) network for AUROC, AUPR, and MCC, stratified by upstream inference backbones (ARACNE, GENIE3, GRNBoost2, Pearson, Spearman, TIGRESS).



Additional Figure A6: AUROC of denoising methods as a function of Abasy *E. coli* snapshot year. Rows correspond to upstream inference methods (Pearson, Spearman, GRNBoost2, GENIE3, TIGRESS, ARACNE); columns correspond to six denoising methods. Panel (B) reports percentage improvement over Base for each denoiser.



Additional Figure A7: AUPR of denoising methods as a function of Abasy *E. coli* snapshot year. Rows correspond to upstream inference methods (Pearson, Spearman, GRNBoost2, GENIE3, TIGRESS, ARACNE); columns correspond to six denoising methods. Panel (B) reports percentage improvement over Base for each denoiser.



Additional Figure A8: MCC of denoising methods as a function of Abasy *E. coli* snapshot year. Rows correspond to upstream inference methods (Pearson, Spearman, GRNBoost2, GENIE3, TIGRESS, ARACNE); columns correspond to six denoising methods. Panel (B) reports percentage improvement over Base for each denoiser.

Project Repository

Code and extended materials including additional experiments and extra figures/tables, are available in the project repository:

<https://github.com/haohaostats/SPSID>